



Figure 3: Cloudify architecture (Source: <http://cloudify.co/guide/3.0/overview-architecture.html>)

by introducing a new ARIA node type for Cloudify, which exposes Project ARIA's capabilities to Cloudify.

 **Note:** ARIA (Agile Reference Implementation of Automation) is a vendor neutral and technology independent implementation of the OASIS TOSCA specification. ARIA offers a command line interface (CLI) to develop and execute TOSCA templates, and an easily consumable software development kit (SDK) for building TOSCA enabled software.

Cloudify Composer and Cloudify UI are the two main frameworks that support integration and unified login; both are interdependent and allow many enhancements.

Cloudify's well-known features are:

- Cloudify supports cloud portability and frees businesses from vendor lock-in, ensuring more flexibility
- It provides a native cloud experience
- It helps in faster app rollout and life cycle automation, shortening the deployment time from days to minutes
- Helps in maximum cloud performance with streamlined processes, enhancing manageability and minimising error
- Helps in complete control, visibility, app-centric activity monitoring
- Cloudify is very cost-efficient and helps to optimise cloud usage
- It helps in application modelling (describes an application with all its resources)

Apart from the above list, orchestration in Cloudify enables maintaining and running identified applications. Pluggability, which is one of the core, unique features of Cloudify, provides reusable components (SDN components, NFV components, and so on) as well as abstraction for the system. And Cloudify's security features control who has permissions to use it to execute various operations.

Cloud Foundry

Cloud Foundry is an open Platform-as-a-Service (PaaS) which provides a choice of clouds, developer frameworks and application services. Cloud Foundry makes it faster and easier

to build, test, deploy and scale applications.

Its features are:

- Application- and services-centric life cycle API
- High performance and dynamic routing
- Data and Web services brokers for cloud brokering
- Linux container management (LXC)
- Features like role based access (RBAC) and teams work well with standards based user authentication and authorisation
- Active application health monitoring and management
- Integrated real-time logging API
- Multi-provider ecosystem

OpenShift

OpenShift is Red Hat's cloud computing PaaS offering. It is an application platform in the cloud where app developers and teams can build, test, deploy and run their applications.

Its features are:

- Built-in support for Node.js, Ruby, Python, PHP, Perl and Java (the standard in today's enterprise world)
- OpenShift provides customisable cartridge functionality for extensions that allow developers to add any other language they wish
- It supports frameworks ranging from Spring and Rails, to Play, etc
- Auto scaling is one of the prominent features of OpenShift; this helps in scaling of applications by adding additional instances
- OpenShift by Red Hat is built on open source technologies (Red Hat Enterprise Linux- RHEL)
- It provides one-click deployment

Tsuru: a PaaS from *Globo.com*

Tsuru is an open source PaaS that was started around January 2012. It supports Mongo, MySQL, Elastic Search, Varnish, Redis, Memcached, Cassandra, etc. Tsuru also supports Go, PHP, Static, Node.js, Java, Python and Ruby platforms for development.

Tsuru can be easily and directly deployed from Git repositories. Scaling up of applications using Tsuru is easy, while its integration with Docker and Kubernetes is also picking up very fast.

Its features are:

- Simple architecture
- Resilience
- Easy customisation
- Completely open source
- Zero downtime 

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